AI-101 MINI PROJECT · TEAM 75

# Prediction of Price of EV

Estimating an electric car’s price from its battery, range and performance specifications

Repository github.com/praveenya1374/ai-project-team-75


THE TEAM

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# The problem and the data

### What we predict

Our dataset lists 10,000 EVs with their specifications and price. We train a model to estimate an EV's price in INR Lakh from those specifications.

### Why it matters

Price depends on battery, range, power and more at the same time. The model shows how much each specification adds to the price.

**10,000**
rows

**9**
columns

**8**
input features

**0**
missing values

**Dataset** india_ev_car_features_price.csv (AI-generated dataset)
**Inputs** Battery capacity, range, motor power, top speed, charging time, seats, fast-charging support, boot space
**Target** Price (INR Lakh)

# Our approach

**1**

## Load data

Read the EV CSV with pandas; inspect shape and types

→

**2**

## Clean

Check for missing values; median fill if any (none found)

→

**3**

## Features

8 specifications as X, price as y

→

**4**

## Train

80/20 split. Multiple Linear Regression on all 8 features

→

**5**

## Evaluate

Predict on 2,000 test cars; MAE, MSE, RMSE, $R^2$

## Libraries used

**pandas**
Data handling

**NumPy**
Numerical work

**Matplotlib**
Charts

**scikit-learn**
Model and metrics

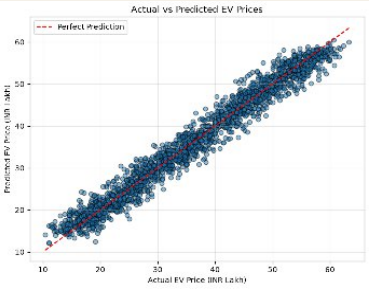
# Results

**0.9665** $R^2$ score

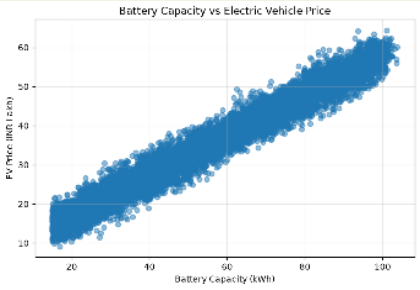
**1.908** MAE (Lakh)

**2.417** RMSE (Lakh)

**5.842** MSE




Actual vs predicted (test set)




Battery capacity vs price

| Feature | Coef. | Feature | Coef. |
|---|---|---|---|
| Battery (kWh) | 0.4136 | Range (km) | 0.0073 |
| Motor power (kW) | 0.0228 | Top speed (km/h) | 0.0110 |
| Charging time (h) | 0.0094 | Seats | 0.2057 |
| Fast charging | 1.2949 | Boot space (L) | 0.0042 |
| Intercept | 1.6086 | | |

In [ ]:

```
# 14.2 Predicting the estimated price

predicted_price = ev_model.predict(new_ev)

print(
    "Estimated Electric Vehicle Price:",
    round(predicted_price[0], 2),
    "INR Lakh"
)
```

```
Estimated Electric Vehicle Price: 38.24 INR Lakh
```

Demo: predicting the price of a new EV (cell 14.2)

Inputs: 60 kWh, 400 km, 150 kW, 150 km/h, 4 h charge, 5 seats, fast charging, 350 L boot

# Learnings

## What worked

- Plain linear regression reached $R^2$ 0.9665
- Battery capacity tracks price almost linearly
- Each kWh of battery adds about 0.41 Lakh; fast charging adds about 1.29 Lakh
- Clean data: no missing values to fix

## What didn't

- One 80/20 split, no cross-validation
- Only a linear model; no other models compared
- Dataset is AI-generated: the high $R^2$ may partly reflect its near-linear pattern, so real prices may differ
- Next: try Random Forest and cross-validate

## Who did what

- Praveenya: dataset loading, exploration, cleaning
- Sriram: features, target, EDA charts
- Aayan: split, training, evaluation
- Ashok: final plots, new-EV prediction, integration, docs, repo

**AI-use disclosure** Our dataset is AI-generated. AI was used only for README formatting and project code structuring. No AI was used to generate the results.